

Original Papers

CASINO : A New Monte Carlo Code in C Language for Electron Beam Interaction —Part I: Description of the Program

PIERRE HOVINGTON, DOMINIQUE DROUIN, RAYNALD GAUVIN

Département de Génie Mécanique, Université de Sherbrooke, Sherbrooke, Québec, Canada

Summary: This paper is a guide to the ANSI standard C code of CASINO program which is a single scattering Monte Carlo SIMulation of electron trajectory in sOLid specially designed for low-beam interaction in a bulk and thin foil. CASINO can be used either on a DOS-based PC or on a UNIX-based workstation. This program uses tabulated Mott elastic cross sections and experimentally determined stopping powers. Function pointers are used for the most essential routine so that different physical models can easily be implemented. CASINO can be used to generate all of the recorded signals (x-rays, secondary, and backscattered) in a scanning electron microscope either as a point analysis, as a linescan, or as an image format, for all the accelerated voltages (0.1–30 kV). As an example of application, it was found that a 20 nm Guinier-Preston Mg_2Si in a light aluminum matrix can, theoretically, be imaged with a microchannel backscattered detector at 5 keV with a beam spot size of 5 nm.

Key words: Monte Carlo simulation, low-energy scanning electron microscope, backscattered electrons, image simulation, line profiles, x-rays $\phi(\rho z)$ computation, Mott elastic cross section, experimental stopping power

1. Introduction

Nowadays, new compounds and alloys are designed to answer specific needs and specific applications. In general, the new high-strength alloys contain a very fine scale microstructure, less than micrometer scale grain size, and precipitates < 0.05 μm . The development of such new materials must be followed by both crystallographic and chemical characterization to yield a full understanding of the microstructure and property relationships. In this perspective, electron microscopy (scanning and transmission) is the most complete and power-

ful instrument that could be used. Scanning electron microscopy (SEM) has the advantage that sample preparation is generally more straightforward compared with the preparation of the electron transparent region needed for the transmission electron microscope (TEM).

Combining a field emission source into an SEM (FESEM) results in a high current density into a very small probe (1 nm @ 30 keV). The FESEM can be used efficiently to characterize the new high-strength alloys as well as complex semiconductor multilayers. In addition, the FESEM opens new opportunities for imaging and microanalysis at low voltage (< 5 kV) such as the examination of non- or poorly conducting materials (Joy and Joy 1996), the imaging dopant concentration in semiconductor (Perovic 1995), quantitative x-ray analysis with resolution < 100 nm, and the imaging of phases < 20 nm (Gauvin *et al.* 1995a). Traditionally, Monte Carlo programs are used to exploit and understand fully the capabilities of electron microscopes (Newbury and Yakowitz 1975) at energy > 10 keV. However, for low energy ($E < 5$ keV), because the first Born approximation is no longer valid, a new generation of Monte Carlo programs is urgently needed. A careful investigation of electron-solid interaction models must be developed before any conclusion can be given.

In this paper, we fully describe a complex Monte Carlo program specifically designed for low-beam interaction that can be used to generate all of the recorded signals (x-rays, secondary and backscattered) in an SEM. This program can also be efficiently used for the range of acceleration voltage found on an FESEM (0.1–30 kV). The goal of this paper is to provide users with a complete description of the most important procedures and features of this Monte Carlo program. Description of the available physics will also be presented. In addition, a full description of the most important routines will be given. This section is more programmer-oriented. As an example of application, the possibility of imaging < 100 nm Mg_2Si precipitates in an aluminum matrix (Al 6061 alloy) will be presented. The sources code and the executable file is now available on the Internet at the address <http://casino.gme.usherb.ca/casino>

2. Features of the Program

This Monte Carlo program contains over 50,000 lines of code in ANSI standard C language distributed over 20 files. It

Address for reprints

Pierre Hovington
Département de Génie Mécanique
Université de Sherbrooke
2500 Boulevard Université
Sherbrooke, Québec
Canada, J1K 2R1

was written to be easily compiled and to run both on a personal computer (PC) and a UNIX workstation with a C/C++ compiler. To increase both the clarity and flexibility of the program, C structures are commonly used for full characterization of all each elements, regions, and electron trajectories (c.f. 3.1). In those structures, all of the necessary data can easily be accessed over the entire program [e.g., for an element: atomic number (Z), mean ionization potential (J), X, Y, and Z position of the electron at the a given collision i, etc. (cf. Table II)].

We use the function pointer for the most essential function (mean ionization calculation, random number generation, total and partial elastic cross section, stopping power, direction cosine, and ionization cross section) so that different physical models can be easily chosen. Those pointers can be assigned to any given or user-defined function. Any subse-

quent calls are made with the function pointer name and will go directly to the assigned function. Hence, experiencing with a different physical model only involves a pointer assignment into the main file and does not require any search and replacement procedure through the entire program.

The Monte Carlo code can be separated into three sections (Fig. 1) : input, main program, and outputs. In the input section, all the necessary variables are entered. These variables are related to the declaration of the specimen nature, the microscope condition, and the choice for the output. The variables can be answered directly from the keyboard through a user-friendly pop-up window or written in a file that can be used as an argument in a given command line structure.

For the specimen, the type of geometry can be chosen among multilayers, vertical plane (or grain boundary), spher-

TABLE I Available output information and extension of the corresponding filename for each output category

Output Category (files format)	Output information available	Output filename extension
Basic simulation information	General information	eba
	• Composition and thickness of all the regions • Geometrical type of the sample	
X-ray generation	Microscope conditions	prz
	• Beam energy • Number of simulated electrons • Beam diameter • Angle between specimen normal and beam direction • Beam position (X and Y coordinate) • Take off angle of the X-ray detector	
Electron distributions	Statistical	rx
	• Backscattered coefficient • Backscattered detector gain • Transmitted coefficient • Mean number of collision • Mean maximum X,Y and Z value reached • Mean maximum depth reached	
Images and Linescans	Distribution	rx
	• $\phi(\rho z)$ of K, LIII and MV shell • $\phi(\rho z)$ of Bremsstrahlung	
Electron distributions	Intensity	rx
	• $K\alpha_1$, $L\alpha_1$ and $M\alpha_1$ lines • Bremsstrahlung	
Electron distributions	Statistical	dzm
	• Mean maximum depth reached • Mean maximum lateral depth reached • Mean maximum lateral depth reached by the Backscattered electrons • Mean number of collision of the Backscattered electrons	
Electron distributions	Energetical	der,det
	• Mean energy of the backscattered and/or transmitted electrons • Mean deposited energy per layer (for multi-layers sample)	
Electron distributions	Fractal	dgl
	• D global (Gauvin <i>et al</i>, 1994)	
Images and Linescans	ASCII	cri
	• Backscattered coefficient at each simulated point • Backscattered detector gain at each simulated point	
Images and Linescans	BINARY (PC)	*ret.bmp
	• BMP image of the backscattered coefficient • BMP image of the backscattered detector gain	

ical inclusion, or step edge. The specimen geometry must be characterized by an analytical equation. These analytical equations are used in conjunction with that characterizing the electron trajectory (straight line) to determine the region change. For multilayers and grain boundary geometry, up to 10 regions can be defined. For each of those regions, the composition and thickness have to be specified. In the case of spherical inclusion, the number of regions is set to two: the inclusion and the surrounding matrix. As for the step edge, the wall angle and the step height can be varied. All of the input information is stored in the regions structure (c.f. 3.1).

The microscope conditions refer to the energy of the primary electrons (beam accelerating voltage), the probe size, the number of simulated electron trajectories, and the angle between the normal specimen and the beam direction.

One important feature of the Main program is that, after each electron trajectory, the entire position of the electron is recorded in a global C structure that can be accessed for any subsequent calculation. In doing this, we ensure that the main interaction loop is kept as simple as possible and that implementation of a new procedure is straightforward.

Several output results can be generated for a given simulation. Using a PC equipped with a standard video graphics adapter (VGA), electrons trajectories can be visualized on a monitor. For the grain boundary, inclusion, and edge, linescans (and images) can be generated. Table I presents, at each simulation point, all the information that can be computed and saved on files within each category. The x-ray and electron distribution files are only created with the user request, whereas the basic simulation information file is created automatically. When x-rays are generated, the intensity file contains the emitted intensity through an angle defined by the take-off angle of the detector of the sample for a beam current of 1 nA. Those intensities must be multiplied by the portion of solid angle of the detector and by the detector efficiency associated with the x-ray line in order to obtain the detected intensity.

Another important aspect about this program is that the most time-consuming procedures have been optimized (Drouin *et al.* 1997, Hovington *et al.* 1997). As an example, data files are only read once and all the necessary values are stored as global variables. Hence, the amount of memory usage depends on the number of elements and regions. However, to reduce the amount of memory needed (especially crit-

ical for a DOS system), dynamic allocations are widely used in the entire program.

3. Description of the Main Program

This section refers to the source code available on the web site “<http://www.gme.usherb.ca/casino>” A more detailed explanation of the program performance is presented. As mentioned earlier, this section is more programmer-oriented since several references are made to the source code.

3.1. Description of the Variables (header.h)

A lot of variables are needed to simulate electron trajectories in solids. All of those parameters need to be organized in the most reliable way to give fast and efficient access to all of the program’s routines. As a consequence, we used a header file containing the global variables organized in structure. This header file “header.h” must be included at the beginning of every file of the project when using a DOS system.

The specimen is described as a group of simple geometry shapes called a “region.” This region is identified by an integer (i) and all the related information is stored in a structure named “Regions[i].” The structure contains the following information (see Table II) : number of elements in the region, array of the corresponding elements, weight fraction of the compounds, atomic fraction of the compounds, density of the region, mean atomic number, and an array describing the geometry of the region. The information regarding an element (atomic number, molecular weight, density, etc.) is stored in a structure named “Elements[i].”

To store the information of an electron trajectory, a structure labeled “Trajet” is used. In this structure, the position at each collision (i) is collected in the corresponding array “X[i]”, “Y[i]”, and “Z[i]” and the remaining energy is put in “E[i].” To keep track of the electron position through the sample, the region number is stored in “Reg[i].” In addition, when the simulation of an electron is completed, the important information is included in the structure to facilitate the computation of the statistical process, for example, the backscattered electron flag (FRetro), the transmitted electron flag (FTrans), the maximum depth of the electron, the mean energy loss, etc. Those structures are declared as global variables, so they are accessible from every part of the program.

Only the principal structure has been described. For further description, interested readers are referred to the comment in the source file.

3.2. Description of the Main File (casino.c)

The main file “casino.c” contains all the declarations of variables and subroutines needed for the program. The definitions of the principal variables have already been explained in section 3.1. In this section, we will focus on the source code of the Monte Carlo main program. Figure 2 shows the flow-chart of the main file “casino.c”. At the beginning, the local

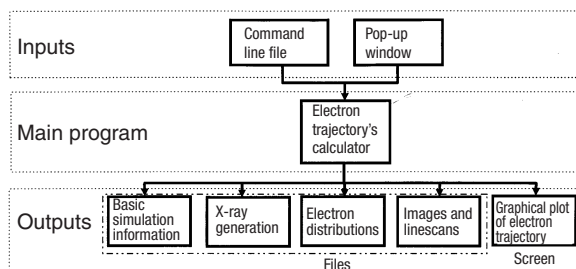


FIG. 1 The three main sections of the CASINO program.

variables and functions are declared. Then, if a batch file argument is passed to the main file from command line, the pointer of the standard input is now pointing to the first data file in the batch file. This means that if the user decides to put all the input data for one simulation in a data file and then writes the name of this data file in a batch file, the program will automatically get the data from this file. When using this line mode command, the program will make a loop until the entire data file is processed (loop 1 in Fig. 2).

As discussed previously (see section 2), this program can easily use different physical models. To achieve this, pointers to subroutines are used. This means that the program will always use the same terminology when calling a routine. As an example, for the total cross section the program calls the routine “Sec_Total()” and it will automatically point at the user’s preferred cross section, for example, “Sec_Total_Mott Fich()”.

The initialization of the specimen and the microscope’s conditions are made through a user-friendly pop-up window. Again, if the program is running from command line, the questions will automatically be answered by the correspond-

ing data in the data file. The call to routine “Initialisation()” makes the memory allocation of the variables needed for the current simulation. This routine also verifies that enough memory is available.

For the same simulation inputs, the program can vary the accelerating voltage from an initial energy (E_{omin}) to a maximum energy (E_{omax}) by a constant step (ΔE_o). This variation is done with the second loop (loop 2 in Fig. 2). To process an image, the beam must scan the surface of the specimen. The discrete position of the beam in the X and Y direction is handled with two loops, and this process is controlled by the structure “Image” (loop 3 in Fig. 2).

When processing data from the Monte Carlo program, it is useful to obtain an estimate of the maximum range reached by the electron for the given simulation we want to perform. We compute this range, at the beginning of any new simulation, from 20 trajectories. Those trajectories are analyzed and the maximum range of the electron (R_{komax2}) is computed. This parameter can be used in further routines.

The principle of all Monte Carlo simulation programs is to generate a large number of events to get good statistics. In the

Table II Description of the structures used by CASINO to simulate each electron trajectory

Description of the structure	Name of the structure	Information contains	Comments
Information related to the description of the region	Regions[i]	Regions[i].NbEl Regions[i].NuEl[] Regions[i].FWt[] Regions[i].FAt[] Regions[i].Rho Regions[i].Zmoy Regions[i].Parametre[]	• Number of element in the region i • Array containing the number of the element in the structure Elements. • Weight fraction of the compound • Atomic fraction of the compound • Density of the region • Mean atomic number of the region i • Array of the parameter describing the region i
Information related to all the elements in the specimen	Elements[i]	Elements[i].Z Elements[i].Nom Elements[i].Rho Elements[i].A Elements[i].J Elements[i].K	• Atomic number • Symbol of the element • Density of the element • Atomic mass of the element • Ionization potential • k parametrization of (Gauvin and L’Esperance 1992)
Information related to the trajectory of the electron	Trajet	Trajet.X[i] Trajet.Y[i] Trajet.Z[i] Trajet.E[i] Trajet.Reg[i] Trajet.FRetro Trajet.FTrans Trajet.FDetec Trajet.NbColl Trajet.Zmax Trajet.LPM Trajet.DedsM Trajet.PhiM Trajet.ThetaM Trajet.XMoyen Trajet.YMoyen Trajet.ZMoyen	• X position of collision i • Y position of collision i • Z position of collision i • Electron energy at collision i • Region number at collision i • Flag of backscattered electron • Flag of transmitted electron • Flag of electron detected in the backscattered detector • Number of collision • Maximum depth • Mean distance between collision • Mean energy lost • Mean azimuthal angle • Mean polar angle • Mean X position • Mean Y position • Mean Z position

case of electron trajectories simulations, a large number of trajectories must be simulated to compute precise results. The choice of the number of electrons simulated (NbElec) depends on the needed precision. A simple Poisson distribution can be used by the user to estimate the error and thus select the appropriate number of electrons (NbElec). To achieve this, a loop is performed to compute the desired number of electron trajectories (loop 4 in Fig. 2). After the simulation of one electron trajectory, all the necessary information is available in the structure “Trajet.” Before the next electron is simulated, the program analyzes this trajectory. For example, the routine “TBaseInter()” is called after each electron trajectory to compute the standard output information (backscattering coefficient, electron range, number of collision, etc.). In this routine each backscattered electron increments the global variable “SCoeffRetro.” When all trajectories are simulated, a final call to the “TBaseFinal()” routine will compute the backscattering coefficient and finally all the outputs will be written in a file with the extension “.eba” (cf. Table I).

The next section will emphasize on the computation of an electron trajectory which is performed in the “Calcul_Trajectoire()” routine.

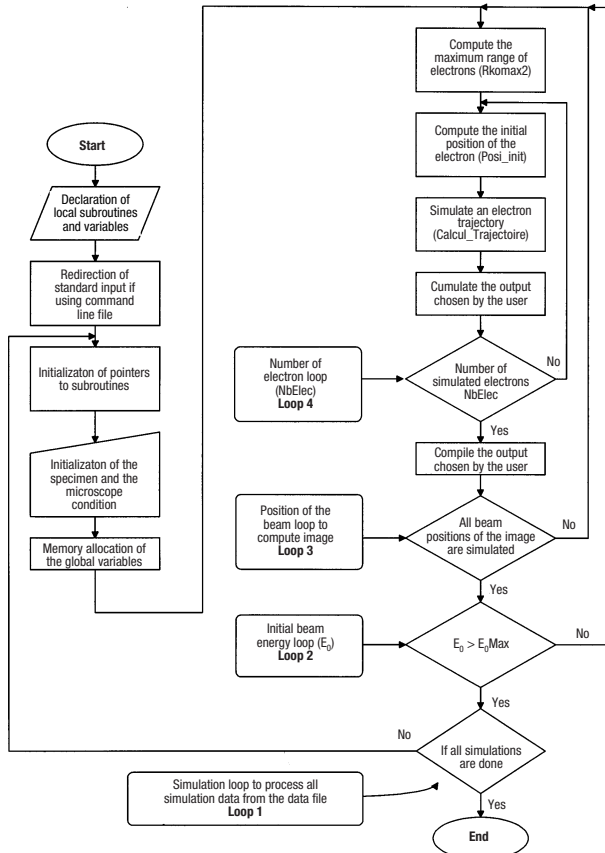


FIG. 2 Flowchart diagram of the main program.

3.3. Description of the Main Routine (Calcul_Trajectoire(), subrou.c)

Figure 3 shows the flowchart of the routine “Calcul_Trajectoire()”. This routine computes an electron’s trajectory taking into account the specimen nature and geometry at a starting position determined by “Posi_Init()”. It will also adjust the starting position of the electron depending on the probe size (BeamDiam). A Gaussian distribution of the electron from the origin of the beam (PositionF_X, PositionF_Y) is used. In addition, the angle between the beam and the sample (Theta0, Phi0) is considered in “Posi_Init()”. The first step is the initialization of variables. In this process, all the variables from the previous simulation are reset to zero. Then the main loop generating the electron trajectory is called. The energy of the primary electron at the current collision “i” is kept in the variable “Energie[i]”. Then the array “SigmaT[]” is initialized with a call to the total cross section “Sec_Total()” for all the elements of the current region. The next step is to determine

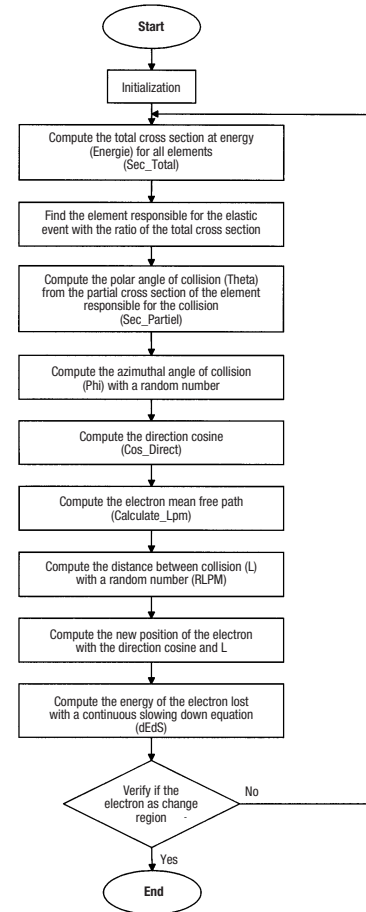


FIG. 3 Flowchart diagram of the “Calcul_Trajectoire()” routine.

which atom is responsible for the elastic scattering. To achieve this, Eq. (1) is used :

$$Random > \sum_{i=1}^n \frac{F_i \sigma_i}{\sum_{j=1}^n F_j \sigma_j} \quad (1)$$

where “Random” is a random number uniformly distributed between 0 and 1, σ_i is the total cross section of element i, F_i is the atomic fraction of element i, and n is the number of elements in the region. When Eq. (1) is true, the element responsible for the collision is i. The polar angle of collision θ “Theta” is determined with the value of the partial cross section “Sec_Partiel” of the element i. This routine computes the polar angle of collision by solving:

$$R = \frac{\int_0^\theta \frac{d\sigma}{d\theta} \sin(\theta) d\theta}{\int_0^\pi \frac{d\sigma}{d\theta} \sin(\theta) d\theta} \quad (2)$$

where $\frac{d\sigma}{d\theta}$ is the partial cross section and R is a random number.

This equation may have a numerical solution if the differential cross section used is simple (Rutherford 1911). However, in the case of the Mott cross section, numerical integration is needed to solve this equation. As a result, the implementation is more complex and either parametrization (Drouin *et al.* 1994a) or tabulation (Drouin *et al.* 1997) of the R value over the scattering polar angle θ must be used. For the program, the return values of “Sec_Partiel()” must be in radian measure from 0 to π .

The azimuthal angle φ “Phi” is uniformly distributed from 0 to 2π and is given by Eq. (3) :

$$\varphi = R \times 2\pi \quad (3)$$

where R is another random number. The φ and θ angles are defined as the angle formed by the last and new directions of the electrons, so we must recalculate the direction relative to a fixed axis. The routine “Cos_Direct()” computes the direction cosine (Rx,Ry,Rz) with the old value (Rx0,Ry0,Rz0).

$$\begin{aligned} Rx &= \frac{Rz0 \times \sin(\theta) \cos(\varphi)}{\sqrt{Rx0^2 \times Rz0^2}} \\ &+ \frac{Rx0 \times Ry0 \times \sin(\theta) \sin(\varphi)}{\sqrt{Rx0^2 \times Ry0^2 + (Rx0^2 + Rz0^2) \times (Rx0^2 + Rz0^2) + Ry0^2 \times Rz0^2}} \\ &+ Rx0 \times \cos(\theta) \\ Ry &= \frac{-\sin(\theta) \sin(\varphi) \times Rx0^2 \times Rz0^2}{\sqrt{Rx0^2 \times Ry0^2 + (Rx0^2 + Rz0^2) \times (Rx0^2 + Rz0^2) + Ry0^2 \times Rz0^2}} \\ &+ Ry0 \times \cos(\theta) \end{aligned}$$

$$\begin{aligned} Rz &= \frac{-Rx0 \times \sin(\theta) \cos(\varphi)}{\sqrt{Rx0^2 \times Rz0^2}} \\ &+ \frac{Ry0 \times Rz0 \times \sin(\theta) \sin(\varphi)}{\sqrt{Rx0^2 \times Ry0^2 + (Rx0^2 + Rz0^2) \times (Rx0^2 + Rz0^2) + Ry0^2 \times Rz0^2}} \\ &+ Rz0 \times \cos(\theta) \end{aligned} \quad (4)$$

This formula is more complex than the one of Soum *et al.* (1979), but we have noticed that their equation does not always satisfy the simple sum rule of cosine :

$$Rx^2 + Ry^2 + Rz^2 = 1 \quad (5)$$

This equation must always be true to compute consistent direction.

To determine the distance (L) between two collisions, Eq. (6) is used :

$$L = -\lambda \log(RLPM) \quad (6)$$

where RLPM is a random number and λ is the electron mean free path. The routine “Calcule_Lpm()” computes the electron mean free path using this equation:

$$\lambda = \frac{1 \times 10^{21}}{No} \frac{\sum_{i=1}^n \frac{C_i A_i}{\rho}}{\sum_{i=1}^n F_i \sigma_i} \quad (nm) \quad (7)$$

where C_i is the weight fraction, F_i is the atomic fraction, A_i is the atomic mass, and σ_i is the total cross section of element i. ρ is the density of the region and No is the Avogadro number. If σ is given in nm^2 , the value of λ will be in nm as expected by the program. Then, with the distance and the direction cosine, the next position is computed and stored in “Trajet” at the position i.

The energy lost during a travel distance “L” is a constant value, since a continuous slowing down approximation is used in this program. The function “dEdS()” computes the rate of energy lost in keV/nm using a user-preferred stopping power equation. The energy at position i is computed by the following equation :

$$E_i = E_{i-1} + \frac{dE}{dS} \times L \quad (8)$$

where E_{i-1} and E_i are the respective energy at previous and current collision and $\frac{dE}{dS}$ is the rate of energy loss.

Before carrying out the next collision, the program will check whether the electron had escaped the region or the specimen. The next section will explain this process in detail.

3.4. Description of the Region Check Routine (Verifi_Region(), subrou.c)

The program uses the routine “Verifi_Region()” to keep track of the electron through all the specimens. This routine has two principal tasks. First, it verifies whether the electron has escaped the specimen through the top or the back of the specimen. Second, it modifies the mean free path in case of a region change.

The first condition verified is the minimum energy: If the energy of the electron is lower than the minimum energy permitted (Eminimum), the program stops the simulation of the electron. Then, if this condition is not satisfied, the program will call a routine that will correct the trajectory depending on the type of region. This routine is unique for each type of specimen. As an example, for a multilayer sample (plane perpendicular to the beam), the routine is “Intersec_RegionF0()”. In this type of routine, the process is separated into two tasks. First, the program calls a function that will initialize two arrays. The array “VecRegion[]” contains all the regions organized in the same order through which the electron travels. The second array “VecFrontiere[]” contains the coordinates of all the intersections of the electron with the regions. Then, with those two arrays, the routine “Correction_Region()” will compute the exact position of the electron according to the change of region.

The routine “Correction_Region()” corrects the path for the region change using the following equation:

$$L_i = L_{i-1} - \lambda_i \times \left(\log(RLPM) + \sum_{j=0}^{i-1} \frac{L_j}{\lambda_j} \right) \quad (9)$$

where λ_i is the mean free path of the region i , RLPM is the same random number used to compute the distance L in Eq.(6), and L_i is the distance traveled by the electron in region i . As this procedure corrects the distance traveled by the electron, it also computes the intersection with the region and stores the coordinate in the structure “Trajet”. If the final region of the electron does not match the region expected in the array “VecRegion[]”, the program restarts the routine “Verifi_Region()” in a recursive scheme.

3.5. Description of Typical Outputs

As mentioned in previous sections, the program is designed for easy incorporation of new output features. In this section a brief description of two routines will be made. The first example is a routine used to calculate the distribution of the maximum range of electrons in a solid. The second routine is related to the generation of x-rays.

3.5.1. Description of the maximum electron depth routine (tdzmax.c): The file “tdzmax.c” contains two routines. The first routine “TDZMaxInter()” is called after each electron trajectory. The second routine “TDZMaxFinal()” is called once all the electrons are simulated. In the first routine, the data are accumulated, and in the second the data are compiled to pro-

duce the desired outputs. This is the principle of all the outputs of the data manipulation routine of the CASINO program.

The initializations of local variables are made at the beginning of “TDZmaxInter()”. Then the program allocates the memory needed for the two arrays “DZMax[]” and “DZMaxRetro[]”. These arrays contain the distributions of the maximum depth of the normal and backscattered electrons, respectively. The routine “IndiceDist()” is used to find the appropriate position in the array of the maximum depth of an electron. This routine needs to know the number of points in the distribution, the minimum and maximum values of the distribution, the value to be indexed (in this case the maximum depth), and the type of distribution (linear or logarithmic). Then, with this index, the appropriate array is increased by one.

Once all the electrons are simulated, the routine “TDZMaxFinal()” is called. In this routine the outputs will be treated and written in a file. Instead of asking the user to enter a filename, the program automatically gets a name from the routine “GetName()”. The first eight characters of this name are the same for all the outputs. This part of the name contains all the information of the specimen and the microscope condition. The first two letters refer to a code that is explained in detail in the data file “NomFichier.Dat.” Each time a simulation is performed, the routine “GetName()” searches in the data file to see whether an identical simulation has already been performed. In the case of a new type of simulation, a new code will be assigned and the simulation parameters will be written in the file “NomFichier.Dat.” The next three characters represent the acceleration voltage used in the simulation. The last three digits are the simulation number used when calculating an image. The three extension letters are used to personalize the name for a specified type of output. For the distribution of the maximum depth, the extension “.dzm” is used. (cf. Table I)

3.5.2. Description of x-ray generation routine (rayonsx2.c): Another example of output of the program is x-ray generation. The two routines to achieve this are “TBaseRayonsX()” and “TPhieRoZRX()”, and they can be found in the file “rayonsx2.c.” The flowchart of the routine “TBaseRayonsX()” is shown in Figure 4. When the routine processes the first electron trajectory, the program will allocate the memory needed for static variables and determine the critical ionization energy for the common shells of all the elements in the specimen.

With the trajectory position stored in the structure “Trajet”, the distance and the mean energy between successive collisions is computed. Then the mean energy is used to calculate the number of ionization events with the ionization cross section “Sec_Eff_Ion()”.

The number of ionization events (n) is stored in an array. The position (i) in the array is determined by the depth (Z) of the ionization event. Those events are assumed to be uniformly distributed between two collisions. The array is divided in constant depth layers (ΔZ) and the number of ionization events (n) is stored in the appropriate position ($i \times \Delta Z = Z$). To process all collisions of the electron trajectory, the

program uses a loop with the number of collision “IndiceColl”. When all the collisions are processed, the routine returns to the main loop and computes another electron trajectory.

The routine “TPhieRoZR(X)” is called once all the electron trajectories are simulated; the corresponding flowchart is shown in Figure 5. This routine will process each element of the specimen individually. At the beginning, the program will determine the critical ionization energy of the common shell (K, LIII, MV) and will store the values in an array. Then, for a single slice of constant depth (ΔZ) in the sample, the number of ionization events (n) will be normalized by the number of ionization events produced by a single film (n_0) of constant thickness (ΔZ). This normalization is applied to all arrays and all shells of elements. After that, the normalized values are printed in an output file with the extension “.prz.”

4. Available Physics

The accuracy of a Monte Carlo code is entirely dependent on the validity of the physical model used. Hence, the user

must be aware of all the physical models available for a given simulation. It is especially important at low energy, where the first Born approximation is no longer valid. Hence, the Mott elastic cross section must be used instead of the conventional Rutherford cross section (Czyzewski *et al.* 1990). In Figure 6, we present the backscattering coefficient of C, Al, Ag, Au from 0.1 to 10 keV computed with Rutherford and Mott (tabulated) cross section. We simulated 250,000 electrons for each point. Clearly, especially for higher atomic number elements, the discrepancy between experimental and simulated backscattering coefficients is more important when Rutherford cross sections are used.

In addition, as the energy is decreasing, the number of inelastic events also decreases. For example, for Si, ionization of the K shell is no longer possible when the electron energy is < 1.84 keV. Hence, the traditional Bethe stopping power expression (Bethe 1933) is no longer valid since the mean ionization potential (J) and the number of effective electrons participating in the excitation up to the energy transfer (Z) must decrease with energy (Hovington *et al.* 1997).

$$\frac{dE}{dS} = -\frac{7.85 \times 10^4 \cdot \rho}{E} \frac{CZ}{A} \ln\left(\frac{1.166E}{J}\right) \quad (\text{keV/cm}) \quad (10)$$

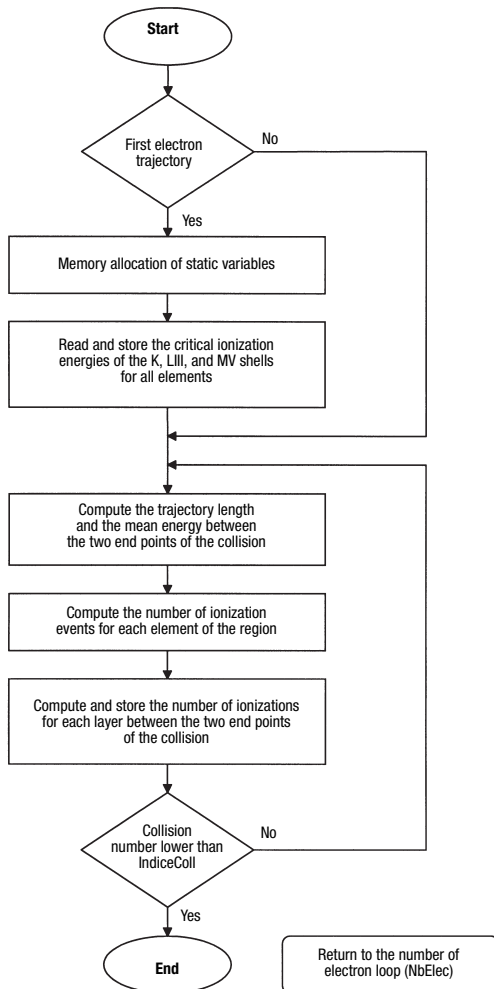


FIG. 4 Flowchart diagram of the “TBaseRayons(X)” routine.

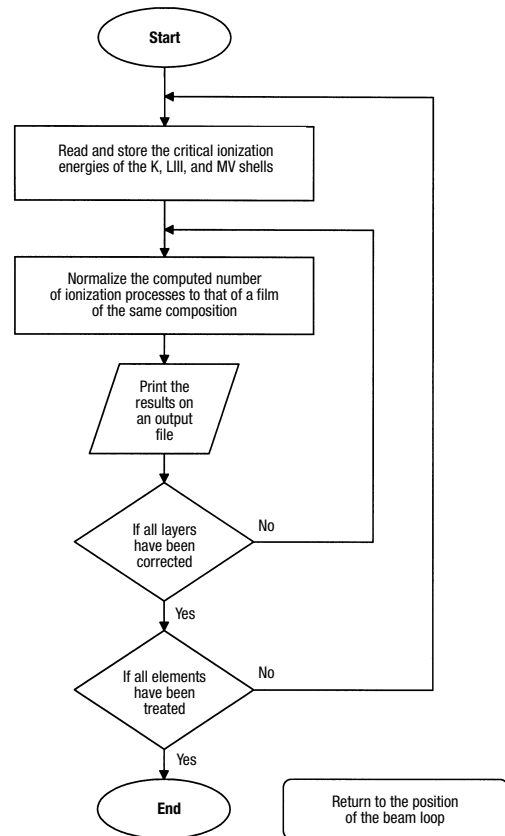


FIG. 5 Flowchart diagram of the “TPhieROZR(X)” routine.

where C and A are, respectively, the weight fraction and the atomic weight (g/mole) of the element. From the Bethe f-sum rule (Bethe 1930), when all the energy loss processes are considered, Z is equal to the atomic number of the element. At low energy, an empirical approach has been suggested by Joy and Luo (1989) who replace the J value by a J^* given by the following expression:

$$J^* = \frac{J}{1 + k \frac{J}{E}} \quad (11)$$

where k is a function of Z (Gauvin and L'Esperance 1992). This modification makes J^* a function of E and removes the problem when $E < J$. However, k values are based only on first-principle calculation and, in addition, the Z value is kept constant with E . For a more accurate modelization, stopping powers computed from experimental measurements are now available on a tabulated form (Hovington *et al.* 1997) or from an analytical expression (Joy *et al.* 1996). Significant differences were observed between the Joy and Luo (1989) expression at very low energy ($E < 1$ keV) and the experimental stopping power for some compounds (Hovington *et al.* 1997).

Hence, this Monte Carlo program contains physical models that are presently considered to be more accurate at low energy (Mott elastic cross section and experimentally determined stopping power).

In addition, we have also incorporated some of the most frequently used physical models. We present in Table III all available physical models for the elastic and ionization cross section, the stopping power, the random number generator, and the x-ray mass absorption coefficient (MAC).

Since Monte Carlo modeling is a statistical process, it is always interesting to investigate the effect of the random number generation on the output results. The system-supplied random generators can sometimes produce correlated and small period random numbers (Press *et al.* 1992). Thus, we implemented three random numbers generators. The period of RAN2 and RAN3 (Press *et al.* 1992) is of 2^{62} compared with 2^{125} for MZRAN13 (Marsaglia and Zaman 1994). However, it should be pointed out that RAN3 is faster than RAN2 and MZRAN13.

The MAC at low energy ($E < 1$ keV) are read from ASCII files (KCOEFF.PRN AND LCOEFF.DON) which contain the tabulated data of Goldstein *et al.* (1984). Those files can easily be read and modified. The modification proposed by Pouchou and Pichoir (1991) has been added. All necessary

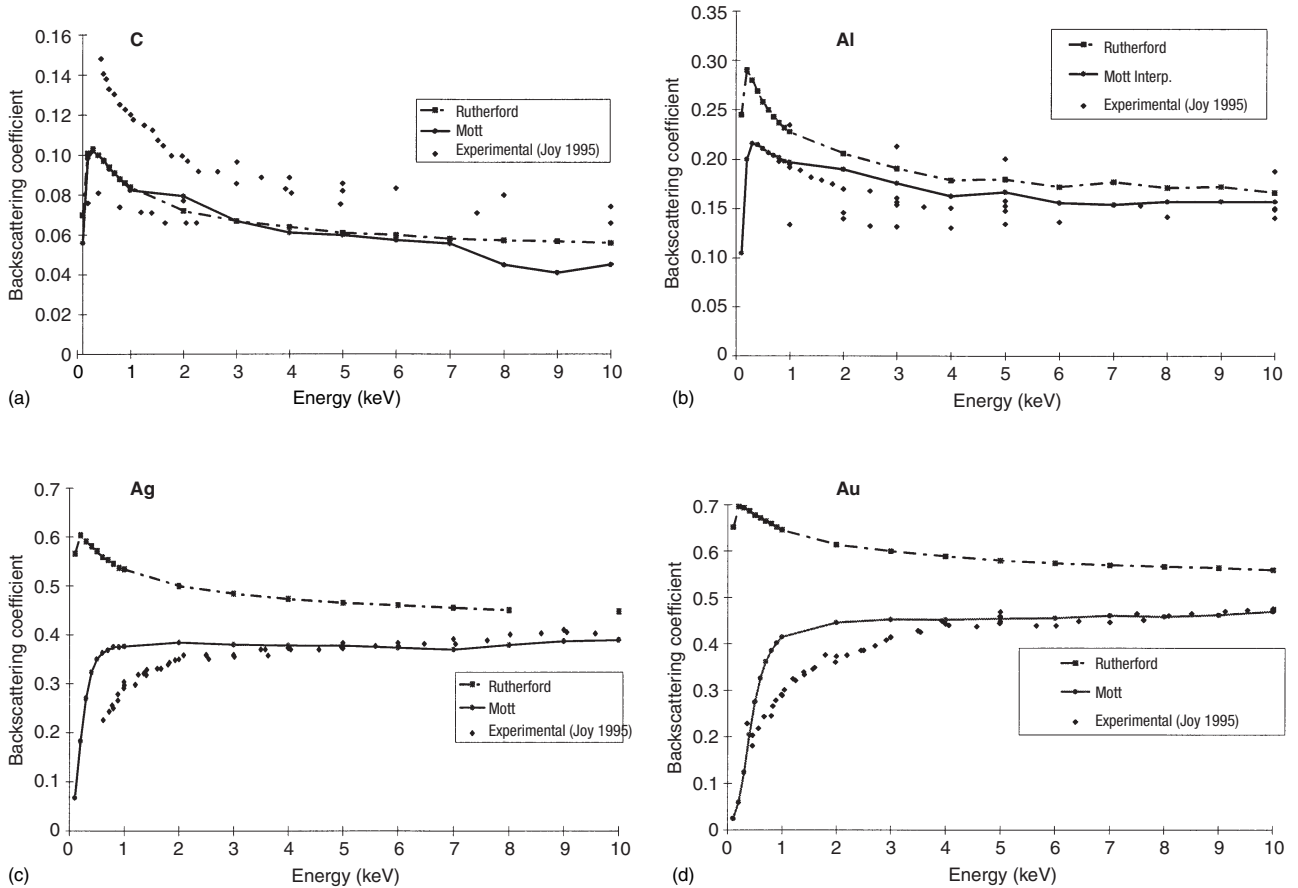


FIG. 6 Backscattering coefficient computed using interpolated Mott cross section, Rutherford cross section, and experimental values compiled in the Joy (Joy 1995) data base. (A) C, (B) Al, (C) Ag, and (D) Au.

energies and weights of shell and subshell are taken from a binary version of the x-ray database available on the MAS Internet web site (<http://www.amc.anl.gov>).

5. Examples of Applications

To illustrate the use of this Monte Carlo program, we investigate the possibilities of imaging (x-ray and backscattered electron) nanometer scale Guinier-Preston precipitates in a light aluminum alloy, such as the commercially available 6061 Al alloy. In this alloy, maximum hardening is reached when the Mg_2Si precipitates are of the order of 10–30 nm (Dorlot *et al.* 1986).

We generate an image from a hypothetical nanometer scale Mg_2Si phase, parallel to the electron beam in an Al alloy. Thus, we model the Al/Si sample with grain boundary type materials containing three regions. The first and third regions contain one element, Al, and the center region contains the Mg_2Si precipitate (Fig. 7). The thickness of the Mg_2Si will vary from 10 to 100 nm.

TABLE III Available physical models for the elastic and ionization cross section, stopping power, random number generator, and the x-ray mass absorption coefficient

Category	Available Models
Elastic cross section	Total • Rutherford (Rutherford, 1911) • Mott (1949) * Analytical parametrization (Gauvin and Drouin, 1993a) • Tabulated (Drouin <i>et al.</i>, 1997) Partial • Rutherford (Rutherford, 1911) • Mott (Mott and Massy, 1949) * Analytical parametrization (Drouin <i>et al.</i> 1994a) * Tabulated (Drouin <i>et al.</i>, 1997)
Ionization cross section	Gryzinsky (1965) Casnati (1982) Gauvin (1993b) Pouchou and Pichoire (1991) Powell (1976) Jakoby (1987)
Stopping power	Bethe (1933) Joy and Luo (1989) Tabulated (Hovington <i>et al.</i> , 1997)
Random number generation	RAN2 (Press <i>et al.</i> , 1992) RAN3 (Press <i>et al.</i> , 1992) MZRAN13 (Marsaglia and Zaman, 1994)
X-ray mass absorption coefficient	Low energy ($E < 1$ keV) • Henke et Eibu (1974) High energy ($E \geq 1$ keV) • Heinrich (1986)

For images, we must specify the X coordinate of the first and last point of the simulation and the total number of points. We set at zero the first interface between the Al and the Mg_2Si . At 1 keV, we choose to simulate 100 points between -50 to 60 nm which gives a separation of 1.1 nm between two simulated points. At 5 keV the interaction volume is larger, so we simulate 100 points between -500 and 500 nm. The microscope conditions are listed in Table IV.

We choose to generate x-ray intensity at each point. In addition, a backscattered image will be created by default. We use an ultra-thin window (Hovington and L'Espérance 1992) energy dispersive spectrometer (EDS) with a take-off angle of 40° and a solid angle of 0.0025Ω to detect the generated x-rays. We set the thickness of both ice and carbon at zero, which corresponds to a detector immediately following a conditioning cycle (Hovington *et al.* 1994, 1995).

We present in Figure 8 the simulated backscattering profiles (linescans) obtained with the 10 and 20 nm Mg_2Si at 1 and 5 keV. With the linescans, we can compute the contrast using the equation of Rosenfield *et al.* (1983):

$$\text{Contrast} = \frac{S_{\max} - S_{\min}}{S_{\max} + S_{\min}} \quad (12)$$

Figure 9 shows the contrast computed at 1 and 5 keV. If we set the minimum observable contrast at 10% (Goldstein *et al.* 1984), we clearly see that, for a 5 keV beam energy, precipitates with a thickness > 20 nm can be detected with a 5 nm probe size, which can be achieved with a FESEM. It is interesting to note that, because of the small difference in the backscattered coefficient at 1 keV between Al and Mg_2Si (0.200 for Al and 0.19 for Mg_2Si), observation of such phases should be performed at 5 keV to obtain the highest observable contrast.

It is important to note that Monte Carlo simulation can be used to investigate the possibilities of the FESEM, but only

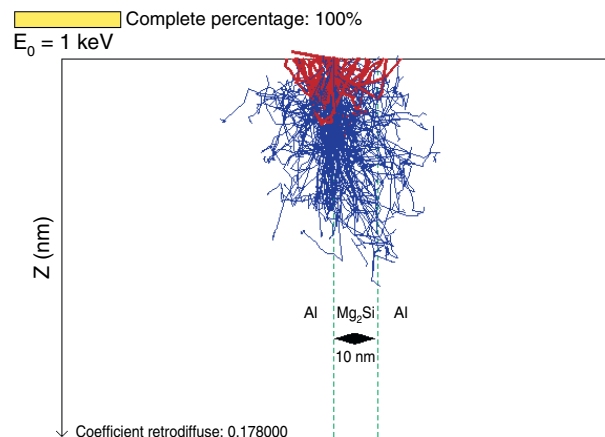


FIG. 7 Typical screen output of electron trajectory. The input parameters are: grain boundary sample with a 10 nm Mg_2Si precipitate between Al analyzed with a beam energy of 1 keV. Beam at the Al/ Mg_2Si interface. The backscattered electrons are plotted in red.

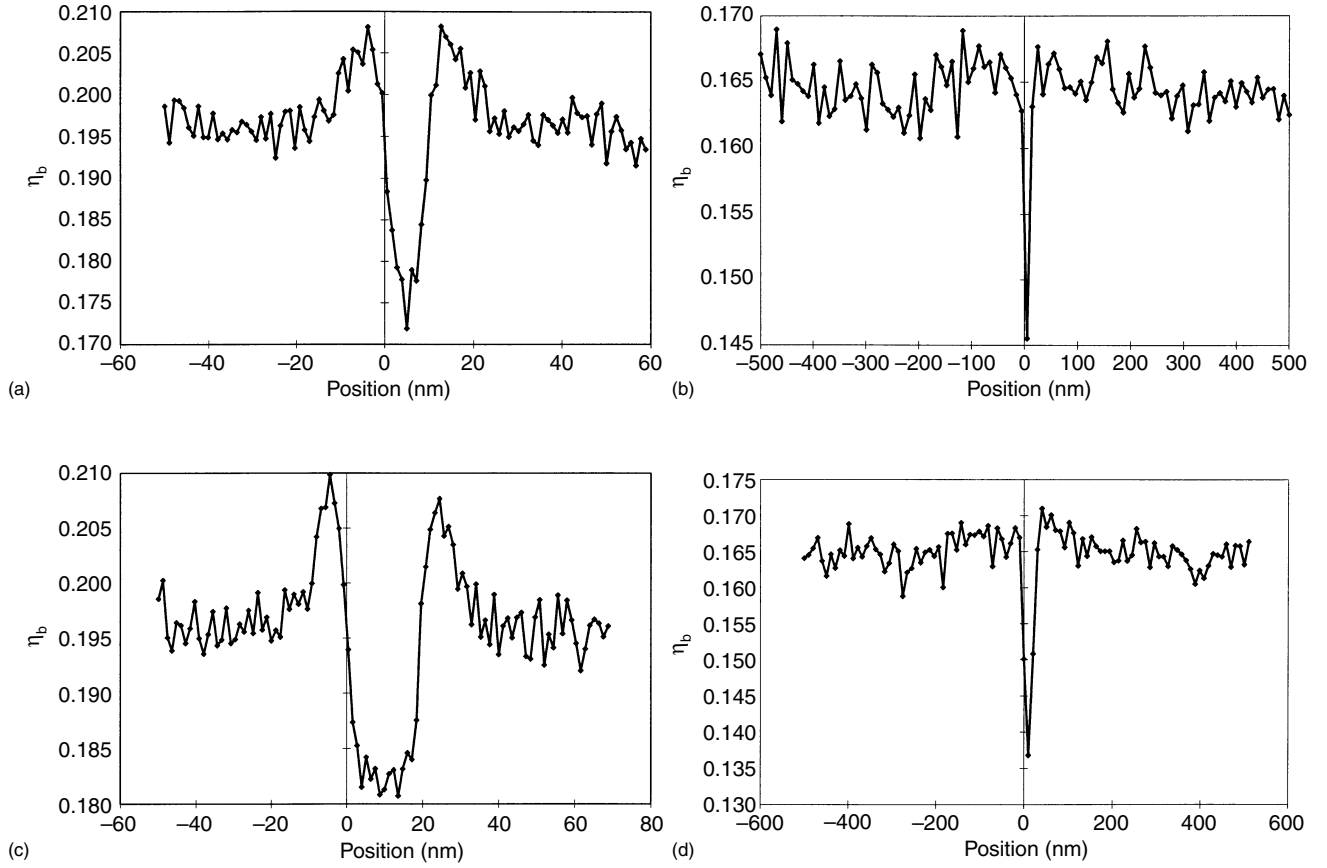


FIG. 8 Backscattered linescan of an Al/Mg₂Si/Al sample. (a) 10 nm Mg₂Si analyzed at 1 keV beam energy, (b) 10 nm Mg₂Si analyzed at 5 keV beam energy, (c) 20 nm Mg₂Si analyzed at 1 keV beam energy, and (d) 20 nm Mg₂Si analyzed at 5 keV beam energy.

experimental results will confirm our prediction, especially at very low accelerating voltage where very few experimental data are available for comparison.

It is fairly easy to obtain from backscattered linescans gray level or color images (Gauvin *et al.* 1995b) which can be more easily interpreted and compared with real images. From those images it is possible to obtain the expected contrast, the measurement errors between the real and the screen (recorded) dimension (Drouin *et al.* 1994b), and the magnification that must be used to image such phases. We present in Figure 10 the gray levels backscattered images of the Mg₂Si precipitate generated with a 5 keV beam energy. The magnification associated with those images is on the order of 300,000 ×.

TABLE IV Microscope simulation parameters used for the Al/Mg₂Si/Al sample

Microscope condition for Guinier-Preston (Mg ₂ Si) phases in an Al/Si alloy	
Accelerating voltage	1 and 5 keV
Probe size	10 and 5 nm respectively
Number of simulated electron trajectory	50 000 per points
Angle between specimen normal and beam direction	0° (normal incidence)

We present in Figure 11 the detected x-ray profile of Al, Mg, and Si K x-ray lines computed with a beam energy of 5 keV on a 10 nm Mg₂Si precipitate. The minimum detectable intensity of both Mg and Si are calculated from a Mg₂Si spectrum generated from first principles (Fig. 12). We use the same experimental conditions and the same ionization cross section as our Monte Carlo program to compute a first-principle spectrum. We compute the minimum detectable intensity (*I*_{min})

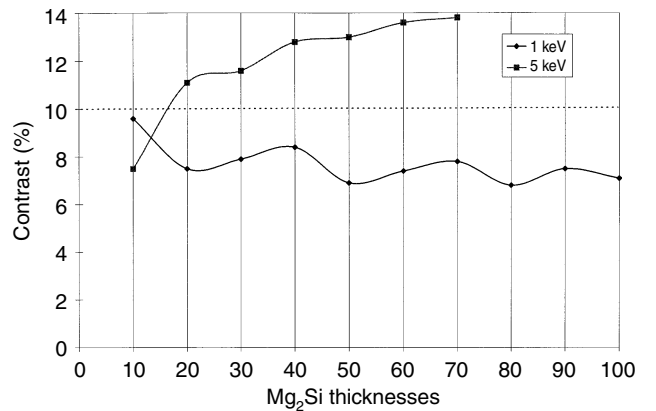


FIG. 9 Computed contrast of an Al/Mg₂Si/Al sample at 1 and 5 keV as a function of the Mg₂Si thicknesses.

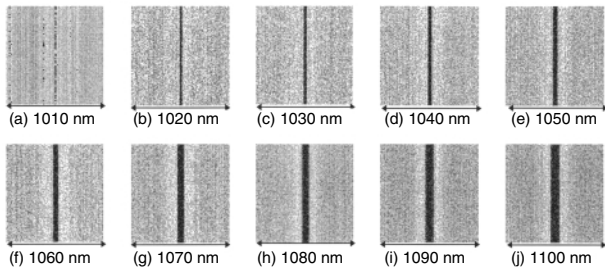


FIG. 10 Backscattered images of the Al/Mg₂Si/Al sample for different Mg₂Si thicknesses. (a) 10 nm, (b) 20 nm, (c) 30 nm, (d) 40 nm, (e) 50 nm, (f) 60 nm, (g) 70 nm, (h) 80 nm, (i) 90 nm, and (j) 100 nm.

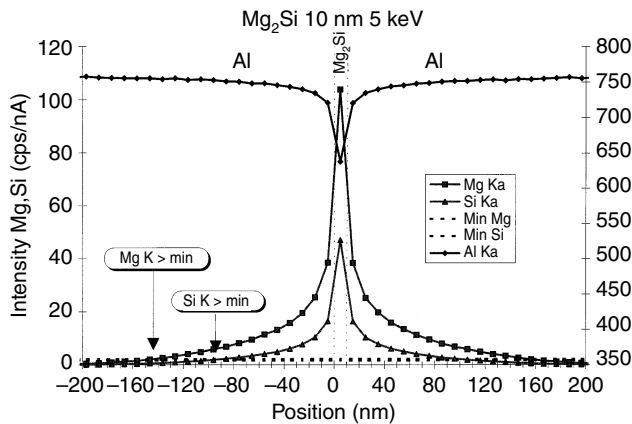


FIG. 11 Detected x-ray profile of Al, Mg, and Si K x-ray lines for a 10 nm Mg₂Si precipitate. The EDS detector parameters are: Si(Li) detector with an Ultra Thin Window (120 nm Al, 100 nm parylene, 50 nm Si dead layer) with a take-off angle of 40° and a solid angle of 0.0025 Ω.

that can be calculated using Eq. (13) (for a confidence interval of 99.9%).

$$I_{\min} = 3\sqrt{2 * I_{BG}} \quad (13)$$

where I_{BG} is the background intensity of the x-ray peak (Fig. 12). From Figure 11, we clearly see that higher spatial resolution is achievable with the Si K line compared with the Mg K as a result of lower excitation overvoltages. However, it is important to point out that a lower detectability limit can be achieved with the Mg K line as a consequence of its higher counts. Clearly, combining generated spectra with Monte Carlo simulation leads to helpful complementary information. The detectability of elements of interest can be efficiently computed before carrying out time-consuming sample preparation and linescans acquisition.

With the same procedure used for computing backscattered images, Figure 13 presents the x-ray elemental color mapping of the Mg K, Al K, and Si K lines for a 10 to 100 nm Mg₂Si precipitate.

We compute the measurement errors by comparing the real (simulated) thickness with the one obtained either from the

full width at half maximum (FWHM) of the Lorentzian fitted to the x-ray linescan or from direct measurement of the simulated backscattering images (Fig. 14). (For images, the thicknesses of the precipitate were determined on the simulated images with a commercial images analysis software (Omnimet 3, Buehler) using two separate techniques. More precisely, the thicknesses were determined from direct measurement on the calibrated BSE images (5 keV) and from a Sobel filter (Russ 1990). The Sobel leads to greater error. Hence, simulation can be used for efficient estimation of the errors when critical measurement must be performed. In addition, when combined with an image analysis software, the proper routine can be predetermined and the associated errors estimated. In this alloy, the thickness of the precipitate is an essential parameter since it determines the mechanical properties of the alloy (Dorlot *et al.* 1986).

6. Conclusion

In the present work, a Monte Carlo program to simulate electron trajectories has been described. It has been demonstrated that the program is useful in many fields of microscopy applications. In addition, this program is specially designed to answer the special problems and needs found at low energy. To achieve this assignment, the program uses the latest available physics to model the electron interaction with the solid. This new model is more complex and, as a result, may need more computing time and a more powerful computer. Hence, the program code has been optimized for speed and portability with more powerful work stations. To show the performance of "CASINO," a speed test was performed. This test consisted of simulating 5,000 trajectories in pure silicon at 1 and 10 keV. The results are shown in Table V. Three codes were tested (Joy 1995, Lowney 1995, and "CASINO"). The first part of Table V gives the results of simulations performed

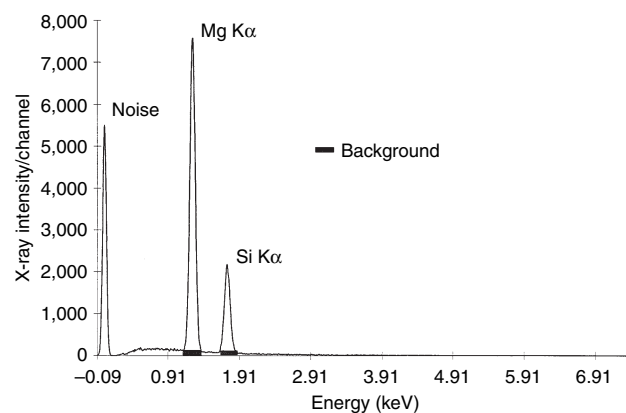


FIG. 12 Mg₂Si spectrum simulated from first principles used to determine the minimum intensity of the Mg K and Si K lines. (100 s acquisition time).

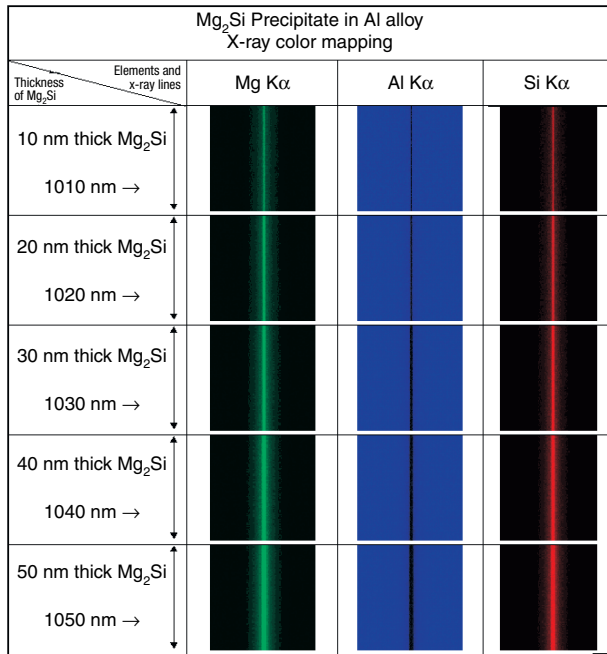


FIG. 13 X-ray elemental color mapping of Mg K α , Al K α , and Si K α for different thicknesses of Mg₂Si in Al/Mg₂Si/Al.

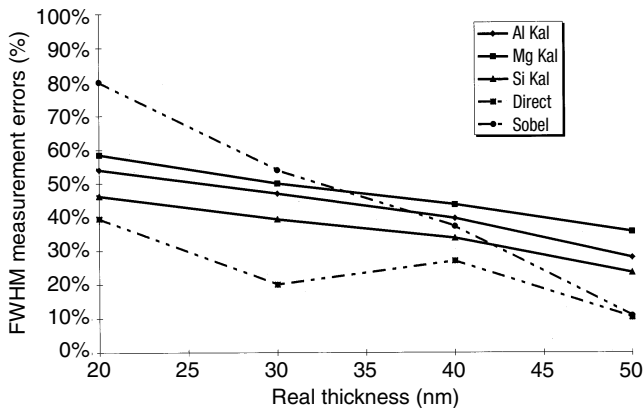


FIG. 14 Errors between simulated and measured Mg₂Si thicknesses as a function of the Mg₂Si thicknesses.

on a 75 MHz Pentium with the electron trajectories display on the screen. In this case, a better performance is achieved with our code against the program of Joy (1995). The second part of the test was carried out on a RISC 6000 model 370 and no graphical output was generated. Again, our program showed better performance, as it is almost 45% faster.

In addition, the program handles multiple specimen geometry for a variety of applications. These features help the user solve many problems often encountered in the electron microscope.

TABLE V Comparison of the processing time needed to simulate 5,000 electron trajectories in Si using three different Monte Carlo programs

	Joy (1995)	“CASINO”
Part I		
(Pentium 75 MHz)	1 keV $\Rightarrow$ 45.6 s	1 keV $\Rightarrow$ 26.4 s
	10 keV $\Rightarrow$ 315 s	10 keV $\Rightarrow$ 175 s
	Lowney (1995)	“CASINO”
Part II		
(RISC 6000)	1 keV $\Rightarrow$ 7.2 s	1 keV $\Rightarrow$ 4.6 s
	10 keV $\Rightarrow$ 57.3 s	10 keV $\Rightarrow$ 39.5 s

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